HATCHSIDE · DESIGN DECISION RECORD

HatchSide kit 1.9 detailed model, corrections

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DOCUMENT CONTROL

Document ID	HTS-DDR-004	Version	0.1
Status	Draft	Date	2026-11-05
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/hatchside		
Source	docs/decisions/0004-kit-1-9-model-corrections.md @ 4004f47		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-05	Amish Chadha	First record, at the kit 1.9.1 upgrade; dimension changes proposed for Amish to confirm

0004: Kit 1.9 detailed model, corrections and dimension changes

- **Date:** 2026-11-05
- **Status:** proposed, awaiting Amish to confirm. The changes are already in `cad/src/model.py` and `bom/bom.csv` because the D3 check cannot pass without them; none changes the pitch, the working principle or a price.

Context

Kit 1.9 requires every option modelled with real stock, every fastener, welds as beads and edge breaks, with no overlapping parts (`python .kit/cad_check.py --export` , 5 mm3 tolerance). The kit 1.7 model of option A used massing shapes for several parts, so building it as a real assembly exposed places where parts overlapped or did not fit. Each is corrected below; the previous design values are kept wherever the check allowed.

Decision

Table 1. Corrections made when the model was rebuilt to D3.

#	Part	Before	Now	Why
K1	Chain shackles	Two screw-pin shackles per chain (BOM line 8)	One 8 mm screw-pin shackle per foot lug, three in all, each joining the ends of two chains; cost unchanged	A shackle cannot join two chains at one lug and clear the lug; the chains meet at the lug
K2	Chain lug and foot plate	Lug 100 x 45 mm, hole 140 mm outboard; foot plate 140 x 280 mm	Lug 44 x 45 mm with the hole at its inner end, still 140 mm outboard; foot plate 140 x 360 mm	The shackle bow must pass the plate end, and the lug must stand on the foot plate; the chain still clears the clevis plates
K3	Head interface	Swivel and shackle packed tight	Shackle depth 36 mm; the interface is 14.2 mm longer	The shackle body clears the swivel eye and the tab with room for the pin
K4	Winch bracket nylon strips	Strips as plain blocks	2 mm nylon strips drilled for the clamp and winch bolts	The bolts passed through solid nylon
K5	Cleat	Placed on the leg	On the inner face of leg B, two M6 through bolts at 40 mm	The cleat and its bolts met the leg wall
K6	Hinge bolts	M16 x 100	M16 x 110 (BOM line 6)	Length for the sleeve, two nylon washers and the nut
K7	Grab pins	Pins placed beside the grab	Hinge, ear and head pins through their holes	The pins were off the part in the old model
K8	Rake	Bails through the top plate	Bails clipped at the plate; drag eye weld legs corrected	Overlap
K9	DomeReach	Pivot across the arm, tiller through the pole	Pivot axis along the swing, ears each side, a split pin, tiller in two halves outside the collar, fork and roller axle with bearing blocks	Overlaps and a missing roller axle; the reach in HTS-CAL-001 section 7 is unchanged
K10	Carry bags	Overlapping solids	Respaced, modelled as 3 mm empty canvas shells	Overlap; the heads are modelled separately
K11	Rim guard splash board	Not riveted	Riveted with 4 mm pop rivets at the stiles	Fasteners must be modelled

Dimension changes to review: K2 (foot plate 80 mm longer, about 0.7 kg each, cost unchanged because galvanizing is priced at a minimum lot) and K3 (interface 14.2 mm longer). All other changes are fastener or placement corrections.

Consequences

- BOM text for lines 1, 6 and 8 changes; no price changes (`budget_usd` stays 5,000).
- HTS-CAL-001 is recomputed from the corrected model: masses 153.0 kg for the whole kit, heaviest load 23.7 kg, tip angle 25.0 degrees; no verdict changes.
- `docs/decisions/0002-design-for-construction.md` is a released record of kit 1.7 values and is not edited; this record supersedes its C6 chain lug dimensions.